3.2.S.4.4 Batch Analysis [2439-exNA, Hongene]

A summary of 2439-exNA drug substance batches manufactured to date is provided in Table 1.

2439-exNA cGMP Batch XXXwas used to manufacture 2439-exNA drug product which is intended for use in proposed Clinical trials. This same batch of drug substance was evaluated in a GLP toxicology study.

Table 1. 2439-exNA Drug Substance Lots Manufactured

Batch Number	Manufacturer	Date of Manufacture	Scale	Batch Size	Purpose/Use
XXX	Hongene	XXX	XXX	XXX	Clinical Batch (drug product manufacture), stability and reference material

The batch analyses of the drug substance lot used in GLP non-clinical and formulation into drug product for proposed clinical studies is provided in Tables 2 and 3.

Table 2. Batch Analysis of antisense single strand

		Batch Number	**XXX**
		Manufacturer	**Hongene**
		Regulatory	**cGMP**
Attribute	**Method**	**Acceptance Criteria**	**Results**
Appearance	Visual inspection	Colorless to pale yellow solution	
Sequencing	ESI-MS/MS	Report sequence coverage	
Denaturing Purity and Impurities	IPRP-HPLC		
Identity	IPRP-HPLC-MS		
Identity	IPRP-HPLC	Retention time consistent with that of reference standard	
Concentration	UV-Visible spectrophotometry	Report value	

Table 3. Batch Analysis of sense single strand

		Batch Number	
		Manufacturer	**Hongene**
		Regulatory	**cGMP**
Attribute	**Method**	**Acceptance Criteria**	**Results**

Appearance	Visual inspection	Colorless to pale yellow solution	
Sequencing	ESI-MS/MS	Report sequence coverage	
Denaturing Purity and Impurities	IPRP-HPLC		
Identity	IPRP-HPLC-MS		
Identity	IPRP-HPLC	Retention time consistent with that of reference standard	
Concentration	UV-Visible spectrophotometry	Report value	

Table 4. Batch Analysis of 2439-exNA Drug Substance

		Batch Number	
		Manufacturer	**Hongene**
		Regulatory	**cGMP**
Attribute	**Method**	**Acceptance Criteria**	**Results**
Appearance	Visual		
Identification (Retention-Time-Non-denaturing)	Non-denaturing SEC		
Identification (Retention-Time -Denaturing)	Denaturing IEX-HPLC	T	

Purity and Impurities	Non-denaturing SEC	Purity Report value	
Purity and Impurities	Denaturing IEX-HPLC		
Tm (thermal melt of duplex)	UV-Visible Spectrophotometry		
Assay[a]	UV-Visible Spectrophotometry		

pH	USP <791>			
Sodium content	Ion Chromatography			
Residual solvents	GC headspace			
Water Content	Karl Fischer			
Bacterial endotoxins	USP <85>			
Bioburden	USP <61>			
Elemental impurities	ICP-MS			